Name

Mathematics Stage 5 Core

Measurement Error

Book 1

Version: 261006

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-MAG-C-01 solves measurement problems by using scientific notation to represent numbers and rounding to a given number of significant figures

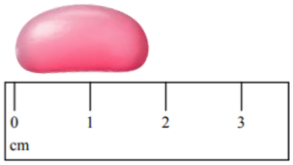
Find absolute and percentage error

- Determine the precision of a measuring instrument by finding the smallest division on the instrument
- Find the absolute error of measuring instruments ($absolute\ error = \frac{1}{2} \times precision$)
- Calculate the percentage error of a given measurement by applying the formula: $\text{percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100\%$

Absolute Error

ⓘ Inaccuracy

- The ruler to the right is measuring the length of a jellybean.
- Since the ruler measures to the nearest centimetre, it is impossible to say precisely what the length is.

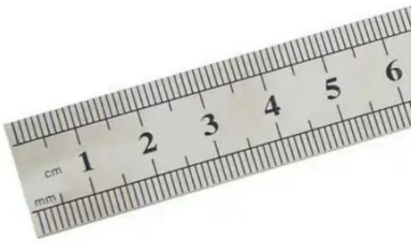


- All measuring instruments have some inaccuracy, depending on how *precise* it is.

ⓘ Precision

- The **precision** of a measuring instrument is the **smallest unit** it can measure.
- Unless otherwise stated, it is the **smallest place value** of the stated measurement.

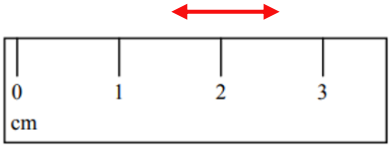
🔍 **Identify** precision of a measuring instrument

a 	**b** 	**c** 
d 18 cm	**e** 2.4 kg	**f** 13.15 L
g 263 ms	**h** 134.7 g	**i** 130 m (nearest 10)
j 1460 g (nearest 10)	**k** 25.50 mm	**l** 3.0 mg

ⓘ Limits of Accuracy

- The **limits of accuracy** describe the possible values that a measurement could be, considering the inaccuracies of the measuring instrument.

On a ruler with 1 cm increments,
a measurement of 2 cm could be anything from 1.5 to 2.5 cm



$$\text{Absolute Error} = \frac{1}{2} \times \text{Precision}$$

$$\text{Lower Bound} = \text{Measurement} - \text{Absolute Error}$$

$$\text{Upper Bound} = \text{Measurement} + \text{Absolute Error}$$

- To find the limits of accuracy:
  1. Identify the precision.
  2. Identify the absolute error.
  3. Find the upper bound.
  4. Find the lower bound.

Example Calculate the upper and lower bound of a measurement

State the limits of accuracy for a measurement of 16 s

1. Precision	1 s
2. Absolute error	0.5 s
3. Upper limit	16 + 0.5 = 16.5 s
4. Lower limit	16 – 0.5 = 15

State the limits of accuracy for a measurement of 9.38 m

1. Precision	0.01 m
2. Absolute error	0.005 m
3. Upper limit	9.38 + 0.005 = 9.385 m
4. Lower limit	9.38 – 0.005 = 9.375 m

Guided Practice

Determine the limits of accuracy for these measurements.

a 72 cm

Precision: 1 cm

Absolute Error: .....

Lower limit: .....

Upper limit: .....

71.5 – 72.5 cm

b 86.6 mm

Precision: .....

Absolute Error: .....

Lower limit: .....

Upper limit: .....

86.55 – 86.65 mm

c 0.57 cm

Precision: .....

Absolute Error: .....

Lower limit: .....

Upper limit: .....

0.565 – 0.575 cm

d 90 m (nearest 10 metres)

Precision: .....

Absolute Error: .....

Lower limit: .....

Upper limit: .....

85 – 95 m

e A rectangular garden was measured as 3 m by 5 m, to the nearest metre.

Find the lower limit of the area of the garden.

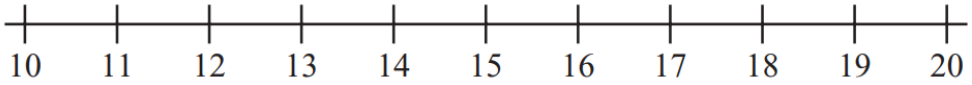


11.25 m²

Key Ideas

1. The precision of a measuring instrument is the ..... unit it can measure,
2. The absolute error is ..... of the precision.
3. The lower bound of a measurement is found by ..... the absolute error from the measurement.
4. The upper bound of a measurement is found by adding the .... to the measurement.

1 Complete the following, referring to the diagram below.



a When using the measurement scale, we are measuring to the nearest ……

1

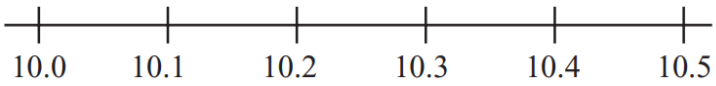
b The absolute error is 0.5 × …… = ……

0.5 × 1 = 0.5

c If a measurement is quoted as 16, it could be anything from …… up to ……

15.5 to 16.5

2 Complete the following, referring to the diagram below.



a When using the measurement scale, we are measuring to the nearest ……

0.1

b The absolute error is 0.5 × …… = ……

0.5 × 0.1 = 0.05

c If a measurement is quoted as 10.4, it could be anything from …….. up to ……

10.35 to 10.45

3 Complete the following for a measurement of 9.45 m

a This measurement of length has been made to the nearest ……,
so the smallest unit on the measuring instrument is ……
Precision = ……

0.01 m, 0.01 m, 0.01 m

b The absolute error = 0.5 × …… = ……

0.005

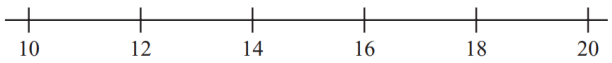
c Limits of accuracy = 9.45 ± …… = …… and ……

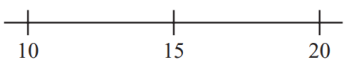
9.45 ± 0.0-5 = 9.445 and 9.455

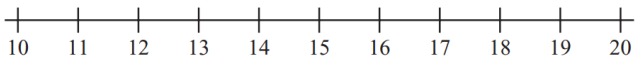
d …… ≤ true length ≤……

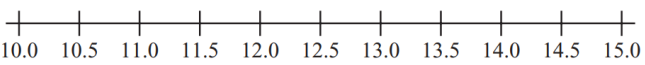
9.445 and 9.455

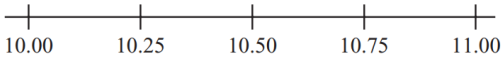
4 Match the scale with the absolute error.

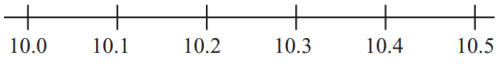










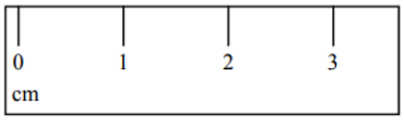


● ±0.125
● ±0.25
● ±0.05
● ±0.5
● ±1
● ±2.5

a – E, b – F, c – D, d – A, e – C

5 Determine the precision and absolute error for these two rulers.

a



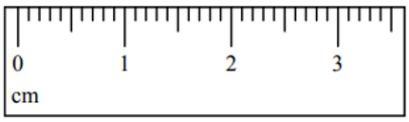
Precision: ……

1

Absolute error: ……

0.5

b



Precision: ……

0.1

Absolute error: ……

0.05

6 The length, x, of a field is given as 90 m.
Write the error interval for x if this measurement is correct to

a the nearest metre

P: 1 m
AE: 0.5 m
$89.5 \leq x \leq 90.5$

b the nearest 10 metres

P: 10 m
AE: 5 m
$85 \leq x \leq 95$

7 Complete the error interval for each statement.

a = 5000 to the nearest integer	☐	$\leqslant a <$ ☐	4999.5 to 5000.5
b = 5000 to the nearest ten	☐	$\leqslant b <$ ☐	4995 to 5005
c = 5000 to the nearest hundred	☐	$\leqslant c <$ ☐	4950 to 5050
d = 5000 to the nearest thousand	☐	$\leqslant d <$ ☐	4500 to 5500

8 Write the limits of accuracy for each of these numbers.

a $p = 60$ to the nearest 10

$55 \leq p \leq 65$

b $q = 24.7$ to 1 decimal place

$24.65 \leq q \leq 24.75$

c $x = 85$, to the nearest integer

$84.5 \leq x \leq 85.5$

d $x = 85.0$, correct to 1 decimal place.

$84.95 \leq x \leq 85.05$

9 The time taken for a plane flight is $6\frac{1}{2}$ hours, to the nearest $\frac{1}{2}$ hour.

Find the lower and upper bounds of the true measurement.

P: 0.5 hours
AE: 0.25 hours
6.75 to 6.25 hours

10 **2019 HSC Standard 2 Band 5**

A person's weight is measured as 79.3 kg. What is the absolute error of this measurement?

P: 0.1 kg

AE: 0.05 kg

11 The size of an angle a is given by the error interval $87.5 \leq a < 88.5°$

a Write three possible values of a correct to 1 decimal place.

87.8, 87.9, 88.3°

b What is the size of angle a correct to the nearest integer?

88°

12 Determine the upper and lower limits of these measurements.

a A section of road to be repaired is measured as 870 m in length, to the nearest 10 m.

865 to 875 m

b A horse is measured to weigh 485 kg, to the nearest 5 kg.

482.5 to 487.5 kg

c A solution is measured to have 4.72 g of copper sulphate, to the nearest 0.01 g.

4.715 to 4.725 g

d A chocolate bar is measured as 91.0 g, to the nearest 0.5 g.

90.75 to 91.25 g

13 Cody measures the mass of an object to be 6 kg.
Jacinta says the same object is 5.8 kg and Lachlan gives his answer as 5.85 kg.

a Explain how all three people could have different answers for the same measurement, assuming they all measured correctly.

They used measurement tools with different precision.

b Write down the possible precision of the measuring instrument being used by each person.

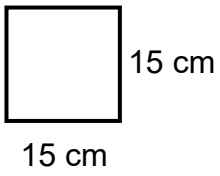
Cody: 1 kg, Jacinta: 0.1 kg, Lachlan: 0.01 kg

14 A square's side length was measured to be 15 cm, to the nearest cm.

a What is the lower limit of the length?

14.5 cm

b Therefore, what is the lower limit of the area?



14.5 × 14.5 = 210.25 cm²

c What is the upper limit of the length?

15.5 cm

d Therefore, what is the upper limit of the area?

15.5 × 15.5 = 240.25 cm²

e Therefore, what are the limits of accuracy for the area?

210.25 to 240.25 cm²

15 Determine the upper limit for the area of this rectangle.



14.5 × 6.5 = 94.25 cm²

16 A rectangular room is 5.4 m long and 3.2 m wide.
What is the lower limit for the area of the room?

5.35 × 3.15 = 16.8525 m²

17 A pizza tray was measured to be 28.6 m in diameter.
What is the lower and upper limit for the area of the pizza tray?

$\text{Lower} = \pi\left(\frac{28.55}{2}\right)^2 = 640.18\ \text{m}^2$

$\text{upper} = \pi\left(\frac{28.65}{2}\right)^2 = 644.72\ \text{m}^2$

Percentage Error

Percentage Error

- The **percentage error** compares the absolute error to the measurement.

$$\text{Percentage Error} = \frac{\text{Absolute Error}}{\text{Measurement}} \times 100$$

1. Identify the precision. This is the smallest place value of the measurement.
2. Identify the absolute error; the absolute error is half the precision.
3. Calculate the percentage error using the formula.

Example Calculate the upper and lower bound of a measurement

18 kg		18.4 kg	
1. Precision	1 kg	1. Precision	0.1 kg
2. Absolute error	0.5 kg	2. Absolute error	0.05 kg
3. Percentage error	$\frac{0.5}{18} \times 100 = 2.778\%$	3. Percentage error	$\frac{0.05}{18.4} \times 100 = 0.272\%$

Guided Practice

Calculate the percentage error of these measurements. Give your answer to 3 d.p.

a 49.6 kg

Precision: ……

Absolute Error: ……

Percentage Error: ……

0.101%

b 251 g

Precision: ……

Absolute Error: ……

Percentage Error: ……

0.199%

c 50 mL (nearest 10 mL)

Precision: ……

Absolute Error: ……

Percentage Error: ……

10%

d 5.256 L

Precision: ……

Absolute Error: ……

Percentage Error: ……

0.010%

Key Ideas

1. The percentage error is the absolute error divided by the ……, then multiplied by 100.

1 Vivian measured the mass of an iPhone as 251 g.

a Identify the precision.

1 g

b Calculate the absolute error.

0.5 g

c Find the limits of accuracy.

250.5 to 251.5 g

d Find the percentage error. Answer correct to two decimal places.

$\frac{0.5}{251} \times 100 = 0.20\%$

2 Abbey measured the mass of a dishwasher as 49.6 kg.

a Find the limits of accuracy.

P: 0.1 kg

AE: 0.05 kg

49.55 to 49.65 kg

b Find the percentage error correct to two decimal places.

$\frac{0.05}{49.6} \times 100 = 0.10\%$

3 Sally measured the mass of an LCD screen as 2.71 kg.

a Find the limits of accuracy.

P: 0.01 kg

AE: 0.005 kg

2.705 to 2.715 kg

b Find the percentage error correct to three decimal places.

$\frac{0.005}{2.71} \times 100 = 0.185\%$

4 Find the percentage error for an object measured as 13.0 cm, to the nearest 0.5 cm.
Answer to 2 d.p.

P: 0.5

AE: 0.25

$\frac{0.25}{13} \times 100 = 1.92\%$

5 2015 HSC Standard 2 Band 5

The length of a fish was measured to be 49 cm, correct to the nearest cm.

What is the percentage error in this measurement, correct to two decimal places?

Precision: 1 cm

AE: 0.5 cm

$$\% \text{ error} = \frac{0.5}{49} \times 100 = 1.02\%$$

6 2014 HSC Standard 2 Band 5

The top of the Sydney Harbour Bridge is measured to be 138.4 m above sea level.

What is the percentage error in this measurement?

P = 0.1 m

AE = 0.05 m

$$\% \text{ error} = \frac{0.05}{138.4} \times 100 = 0.036\%$$

7 2008 HSC Standard 2 Band 4

The capacity of a bottle is measured as 1.25 litres correct to the nearest 10 millilitres.

What is the percentage error for this measurement?

Precision: 10 mL

AE: 5 mL

$$\% \text{ error} = \frac{5 \text{ mL}}{1250 \text{ mL}} \times 100 = 0.4\%$$

8 2017 HSC Standard 2 Band 5

The length of a netball court is measured to be 30.50 metres, correct to the nearest centimetre.

What is the lower limit for the length of the netball court?

Precision: 1 cm

AE: 0.5 cm

Lower limit = 30.5 m – 0.5 cm = 30.495 m